

Forensic Report

Note

Analysis of Disk Image

Overview of mmls Results

The mmls file is located at `C:\Program Files\Foxit Software\Foxit Reader\Foxit Reader.exe`. The file is a FAT16 file, and the mmls file is located at `C:\Program Files\Foxit Software\Foxit Reader\Foxit Reader.exe`. The file is a FAT16 file, and the mmls file is located at `C:\Program Files\Foxit Software\Foxit Reader\Foxit Reader.exe`.

Disk Structure Analysis

Partition Table Information

- DOS Partition Table
- Offset Sector: 0
- Units are in 512-byte sectors

Partition Details

1. Meta - Primary Table (#0)

- Start 0
- End 0
- Length 1
- Description

2. Unallocated (0000000000 - 0000000031)

- Length 32
- Description

3. DOS FAT16 (0x04) (0000000032 - 0000061439)

- Length ,40
- Description 1 1

4. NTFS / exFAT (0x07) (0000061440 - 0000124927)

- Length ,4
- Description
 - NTFS
 - exFAT

5. Unallocated (0000124928 - 0000133119)

- Length ,1 2
- Description

6. Meta - DOS Extended (0x05)

- Length 0
- Description

7. Meta - Extended Table (#1)

- Start 00001310 2
- End 00001310 2
- Length 1
- Description

8. Linux (0x83) (0000133120 - 0000202751)

- Length 32
- Description 0 3.

9. Unallocated (0000202752 - 0000204799)

- Length 2,04
- Description

Anomalous Observations

- Large Unallocated Spaces

Recommended Next Steps

1. Forensic Imaging
2. Data Recovery Tools
3. Partition Analysis
4. File Carving
- Regular Audits

Analysis of Disk Partition Statistics

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First File System: FAT16

File System Information

- File System Type 1
 - FAT16
- OEM Name .0
 - .0.
- Volume ID 0 44
 -
- Volume Label (Boot Sector)

- **Boot Sector**
 - 1
- **Volume Label (Root Directory)** 14 22 1
 - 1
- **File System Type Label** 1
 - 1
- **Sectors before file system** 32
 - 1

File System Layout (in sectors)

- **Total Range** 0 - 140
 -
- **Reserved** 0 - 3 & **Boot Sector** 0
 -
- **FAT 0:** 4 - 241, **FAT 1:** 242 - 479
 -
- **Data Area:** 480 - 61407
 -
- **Root Directory** 4 0 - 11
 -
- **Cluster Area** 12 - 140
 -

Metadata Information

- **Range** 2 - 4 4
 - [Range 2](#), [Range 3](#), [Range 4](#)
- **Root Directory** 2
 - [Root Directory](#)

Content Information

- **Sector Size** 12, **Cluster Size** 12
 - **Sector Size** 12, **cluster size** 12
- **Total Cluster Range** 2 - 0
 - **Cluster Range** 2 - 0

FAT Contents (in sectors)

- -
 - **EOF**
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Suspicious Observations for FAT16

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Recommended Next Steps for FAT16

- "14 22 1"- , ,
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Second File System: NTFS

File System Information

- File System Type ()
 - ,
- Volume Serial Number 0 34 334 3
 - 1
- OEM Name
 - -
- Volume Name
 -
- Version
 -

Metadata Information

- First Cluster of MFT 10 1 & MFT Mirror 1 1
 - Master File Table (MFT) -
 - MFT Mirror ,
- Size of MFT Entries 1024 & Index Records 40
 -
- Range 0 - 44 , Root Directory
 -

Content Information

- Sector Size 12, Cluster Size 1024
 - ,
- Total Cluster Range 0 - 31 42, Total Sector Range 0 - 34
 -

\$AttrDef Attribute Values

- (\$)
- Resident
- Non-resident
- Index

Suspicious Observations for NTFS

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Recommended Next Steps for NTFS

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Third File System: Ext3

File System Information

- **File System Type** 3 ()
 -
- **Volume Name**
 -
- **Volume ID** 2 4 1 4 3321 3
 -
- **Last Written at, Last Checked, Last Mounted at** 2010
 - **Unmounted Properly**
- **Mounted on** /
 -
- **Source OS**
 -

Metadata Information

- **Inode Range** 1 - 0
 - **Inode**
- **Root Directory** 2
 -
- **Free Inodes** 3
 -

Content Information

- **Block Range** 0 - 1 40 , **Block Size** 204
 -
- **Free Blocks** 433
 -

Block Group Information

- - **Number of Block Groups** 2
 - **Inodes/Blocks per group**
 - **Free Inodes/Blocks**

Suspicious Observations for Ext3

- **Volume Name** ,

Recommended Next Steps for Ext3

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Analysis of File Listings

Understanding the File Listings

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- l/l
- realloc

Metadata Information

MFT

- **\$MFT (Master File Table)**

Cluster Size and Unallocated Space

- **Cluster**
- **Unallocated Space**

Key Observations

Application and System Files

Software Identified

- **Android Forensics Directory**
- **Ericom Blaze**
- **Downloadable Executables**

Programming and Source Code

Suspicious or Anomalous Entries

Reallocated Files

- **Reglookup Guide.desktop, Scalpel Data Carving Guide.desktop, Sleuthkit Informer Archive.desktop,**

Deleted Entries

- $$\bullet \quad \text{if } \mathcal{A} \text{ is a } \mathcal{C} \text{ then } \mathcal{A} \text{ is a } \mathcal{C} \text{ (} \mathcal{A} \text{ is a } \mathcal{C} \text{)} \quad (*) \quad (\text{if } \mathcal{A} \text{ is a } \mathcal{C} \text{ then } \mathcal{A} \text{ is a } \mathcal{C}),$$

System Directories and System Files

- **System Volume Information** \$Secure
- \$LogFile, \$Bitmap

Recommendations for Next Steps

1. **Deeper File Content Analysis**
• **File Signature Verification** (e.g., PE, ZIP, PDF, etc.)
• **Metadata Extraction** (e.g., creation/modification dates, author, etc.)
2. **Timeline Analysis**
• **File System Forensics** (e.g., NTFS, FAT, etc.)
• **Registry Analysis** (e.g., Windows Registry, etc.)
3. **Unallocated Space Examination**
• **File System Forensics** (e.g., NTFS, FAT, etc.)
• **Memory Forensics** (e.g., RAM, etc.)
4. **Scanning for Malware**
• **User Activity Profiling** (e.g., network logs, etc.)
• **Backup and Imaging** (e.g., disk, etc.)
• **Log Review** (e.g., system logs, etc.)
• **Expert Collaboration** (e.g., forensic analysts, etc.)

Analysis of Images Found

Overview of Forensic Tools and Concepts

- **File System Forensics** (e.g., NTFS, FAT, etc.)
• **Memory Forensics** (e.g., RAM, etc.)
• **icat Function** (e.g., `icat -i /path/to/image`)
• **Metadata** (e.g., creation/modification dates, author, etc.)
• **File Signature Verification** (e.g., PE, ZIP, PDF, etc.)
• **Timeline Analysis** (e.g., file system forensics, registry analysis, etc.)
• **User Activity Profiling** (e.g., network logs, etc.)
• **Backup and Imaging** (e.g., disk, etc.)
• **Log Review** (e.g., system logs, etc.)
• **Expert Collaboration** (e.g., forensic analysts, etc.)

Image Analysis

1. Filename: /tmp/pet.jpg

- **Description**
• **Analysis**
 - **Contents Significance**
 - **Anomaly Flag**

2. Filename: /tmp/snowy.jpg

- **Description**
- **Analysis**

- Contents Significance
The file's content is highly relevant to the investigation, as it appears to be a screenshot of a web application interface, possibly a login or registration page, which could contain sensitive information or user credentials.
- Anomaly Flag
The file's location in the /tmp directory is unusual, as it is typically used for temporary files. This suggests the file may have been created or modified by a process running on the system, which could be a point of interest for further investigation.

3. Filename: /tmp/webmin.jpg

- Description
The file is a screenshot of a web application interface, likely a login or registration page, showing a form with fields for username and password, and a submit button. The interface is rendered in a browser window, and the file is named webmin.jpg.
- Analysis
 - Contents Significance
The file's content is highly relevant to the investigation, as it appears to be a screenshot of a web application interface, possibly a login or registration page, which could contain sensitive information or user credentials.
 - Anomaly Flag
The file's location in the /tmp directory is unusual, as it is typically used for temporary files. This suggests the file may have been created or modified by a process running on the system, which could be a point of interest for further investigation.

Recommended Next Steps

1. Contextual Verification
 - Verify the file's location and permissions, ensuring it is not a system file or a file owned by a user with elevated privileges.
 - Check the file's metadata, including the creation and modification dates, to determine if it is a recent file.
2. Branding Usage Audit
Conduct a thorough audit of the file's content, focusing on identifying any branding or logos that may be associated with the web application. This will help determine the application's identity and potentially its security features.
3. Content Authenticity
Verify the file's content against the original source, ensuring it is a genuine screenshot of the web application and not a manipulated or altered version.
4. Metadata Retrieval
Retrieve the file's metadata, including the creation and modification dates, to determine if it is a recent file. This information can be used to correlate the file with other evidence in the investigation.
5. Forensics Consistency Check
Perform a consistency check on the file's content, ensuring it matches the original source and is not a manipulated or altered version. This will help determine the file's authenticity and its potential value as evidence.

Forensic Analysis Conclusion

Findings

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Disk Structure and File Systems

- Partition Structure
The disk structure is consistent with the findings, showing a clear separation between the operating system and the user data. This suggests the file was created or modified by a process running on the system, which could be a point of interest for further investigation.

Recommended Actions

1. **Forensic Disk Imaging**
2. **Advanced Data Recovery**
3. **Volume and Label Analysis**
4. **Permission and Access Scrutiny**
5. **Timeline Construction**
6. **Image Authenticity Verification**
7. **Comprehensive Malware Evaluation**

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